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Rate of Force Development Adaptations After Weightlifting-Style Training: The Influence of Power Clean Ability

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Abstract

James, LP, Suchomel, TJ, Comfort, P, Haff, GG, and Connick, M. Rate of force development adaptations after weightlifting-style training: the influence of power clean ability. *J Strength Cond Res* 36(6): 1560–1567, 2022—This experiment examined changes to the rate of force development (RFD) expressed under loaded jump conditions between individuals with a higher (stronger) and lower (weaker) weightlifting performance (as assessed by the 1 repetition maximum [RM] power clean) after training with the weightlifting derivatives. Two groups of markedly different weightlifting ability undertook 10 weeks of training with the power clean variants, snatch pulls, and jump squats across heavy and light conditions. Testing was performed at baseline, after 5 weeks of training (mid-test) and after training (post-test). During testing, RFD was assessed under a series of loads (20–80% squat 1RM) through the jump squat. Furthermore, the force-velocity relationship, and unloaded jump strategy (through the force-time curve waveform), were also examined. Very large change (Hedge's *g*, 95% confidence interval [g] = 2.10, 1.24 to 4.16) in RFD at 20% 1RM at mid-test occurred within the stronger group. Conversely, a small increase ($g = 0.27$, 0.53–1.91) among the weaker subjects existed in this measure at mid-test, reaching a moderate increase at post-test ($g = 0.71$, −0.18 to 2.15). Limited improvements were seen by the stronger subjects in RFD at 60 and 80% 1RM at either mid-test (60%: $g = 0.27$, −0.75 to 1.33; 80% = 0.02, −1.01 to 1.00) or post-test (60%: $g = 0.52$, −0.38 to 1.80; 80% = −0.26, −1.23 to 0.77). The stronger group experienced a shift throughout the force-velocity relationship while a more force-dominant adaptation occurred in weaker subjects. Differences in jump strategy between groups were also noted. Such training will elicit practically different adaptations in rapid force production depending on the individual's baseline weightlifting ability.

Key Words: strength training, neuromuscular, force-velocity, jump squat, countermovement jump

Introduction

The ability to produce high rates of force development (RFDs) is considered a vital component of athletic performance (22). Similarly, many sporting actions require forces to be expressed with maximal intent over a finite period (1,28), generating an impulse that causes a change in momentum of an athlete, their opponent, the athlete-opponent system, or an implement. In sport, expressions of both maximal RFD and maximal impulse often occur throughout the loading spectrum ranging from no external load to events where an athlete must resist forces applied by opponents (27). This is particularly apparent in collision sports where both high force and high velocity expressions drive performance (3,15,16). For example, when striking, sprinting, or jumping, an athlete must apply force only against their bodyweight (their mass and gravity). However, during collisions, such as tackles and grappling encounters, there are additional external forces to resist. For this reason, the enhancement of RFD and impulse under a range of force-velocity conditions is of great relevance to those responsible for the physical preparation of athletes.

A number of resistance training modalities can be used to develop the qualities of RFD and impulse, including heavy strength training, ballistic tasks, and plyometric exercises. Weightlifting derivatives are also considered an effective approach (24,25), with evidence of superior strength-power adaptations when compared with alternate approaches such as jump training and traditional resistance training (12,29). This is likely due to their ability to provide an overload stimulus during the triple extension of the knee, hip, and ankle joints through combined high force and high velocity demands, alongside variation in the type of triple extension pattern that can be used (i.e., different contribution from hip versus knee extensors) (25).

Although the aforementioned training modalities can result in strength-power improvements, the extent and nature of adaptation seems to differ based on pre-existing physical qualities in some, but not all cases. A previous investigation (18) demonstrated that individuals with a 1 repetition maximum (1RM) squat of 2.0 × body mass (BM) produced significantly greater improvements in unloaded countermovement jump (CMJ) peak velocity when compared with individuals with a 1RM squat of 1.2 × BM. Similarly, those with

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greater power clean ability tend to improve unloaded CMJ RFD earlier after initiating combined heavy and light ballistic training (17). However, an investigation by Wilson et al. (30) noted no difference in adaptations to unloaded vertical jump height or flying 20-m sprint time between much stronger and weaker subjects after plyometric training. As such, the impact of baseline physical characteristics on training adaptations, particularly measures of RFD and impulse under heavier loads, is still unclear.

Neuromuscular adaptations including alterations to the pattern of force application (jump strategy) and the force-velocity relationship are believed to underpin rapid expressions of force in sports-specific tasks (8,9). Indeed, previously researchers have shown significant changes to the CMJ force-time waveform following strength and power training (8,18). In addition, analysis of the whole-body force-velocity relationship derived from the incremental load jump squat has detected training-induced changes in the intact neuromuscular system's ability to produce force during an athletic movement against external resistances (8,18). As such, these procedures provide insight into the mechanisms driving changes in explosive performance.

Because of the paucity of research exploring changes to loaded RFD and impulse after training, and the impact of strength level on such changes, the purpose of this experiment was to determine whether adaptations in these measures differed between individuals with a higher and lower weightlifting performance (as assessed by the 1RM power clean) after training with the weightlifting derivatives. A secondary objective was to compare changes to jump strategy and the jump squat force-velocity relationship between the two groups.

Methods

Experimental Approach to the Problem

Subjects were stratified into a stronger or weaker group based on their 1RM power clean result. The experiment lasted 14 weeks including familiarization, testing, and 10 weeks of resistance training. Testing was performed at baseline, after 5 weeks of training (mid-test) and after training (post-test). During testing, RFD, impulse, and peak force were assessed under a series of loads. Furthermore, the force-velocity relationship and jump strategy were also examined.

Subjects

Twenty recreationally trained men with a diverse strength level undertook all training and testing requirements of this investigation. Subjects were ordered based on their 1RM power clean result, after which data were then removed from the middle 4 subjects enabling the comparison of 2 groups with markedly different power clean ability. Data from the remaining 16 men were then used for this experiment (stronger: 24.1 y [SD = 5.1]; weaker: 25.9 y [SD = 2.6]; stronger: n = 8; BM = 78.1 ± 4.0 kg; height = 1.74 m; 1RM power clean = 1.08 ± 0.09 kg·kg^{-1}; 1RM squat = 2.0 ± 0.2 kg·kg^{-1}; resistance training experience = 3.5 ± 1.5 years; weaker n = 8; BM = 82.6 ± 14.0 kg; height = 1.81 m; 1RM power clean = 0.78 ± 0.1 kg·kg^{-1}; 1RM squat = 1.38 ± 0.32 kg·kg^{-1}; resistance training experience = 1.4 ± 1.5 years). All subjects were 18 years or older, provided written informed consent, and were made aware of the risks associated. This study was approved by the Bellberry Human Research Ethics Committee.

Procedures

Training Program. Before the commencement of training and testing, all subjects participated in three 1-hour familiarization sessions covering all training and testing requirements. The training intervention involved 3 sessions per week, separated by a minimum of 24 hours, across 10 weeks of training. Training was divided into two 5-week mesocycles partitioned by a 1-week restitution microcycle during which only mid-testing requirements were undertaken. Throughout the initial 5 weeks of training the first and third workout consisted of the power clean at 70% 1RM, in addition to the jump squat with 40–50% of the 1RM squat. Day 2 consisted of the hang power clean and snatch grip pull performed at 55 and 70% of the power clean 1RM, respectively. Five sets of 5 repetitions and 4 sets of 5 repetitions were undertaken on day 1 and 3, and day 2, respectively. In the second 5-week mesocycle, the power clean was increased to 85% 1RM for 4 repetitions per set, while the jump squat was reduced to 0–30% of the 1RM squat. The hang power clean and snatch grip pull were increased to 70 and 85% of the power clean 1RM, respectively (at 4 repetitions per set). A depth jump was added to day 1 and 3, while a plyometric split squat was added to day 2. Both the 1RM power clean and 1RM squat were reassessed at mid-test (using established protocols (18)) and training loads adjusted accordingly. All training sessions were supervised and commenced with a general, then specific dynamic warm-up. A series of warm-up sets at incremental loads was also undertaken for each exercise. Subjects were asked to refrain from any additional lower body training for the duration of the experiment.

Testing Overview. Baseline testing occurred at least 48 hours after the final instructional session, while 3–5 days and 7–10 days of recovery was enforced before mid-test and post-test, respectively. Testing sessions commenced with the 1RM squat to enable jump squat testing and training conditions to be established. The jump squat at incremental loads was performed while simultaneous force platform readings were acquired. Experimental procedures are described in detail below.

Jump Squat. At least 15 minutes after completion of the 1RM squat using previously reported procedures (18), jump squats were undertaken at progressively increasing, nonrandomized loads based on the individuals' 1RM squat (unloaded, +20%, +40%, +60%, +80% 1RM). All jump squats were performed with a countermovement to a depth resulting in an internal knee angle of 85°. Subjects were instructed to jump for maximal height. A minimum of 2 trials were undertaken with 3 minutes of passive recovery between each attempt. All jump squats were executed on a force platform (Bertec Corporation, Columbus, OH) with the analogue signal sampled at 2,000 Hz using a data acquisition device (NI USB-6259 BNC; National Instruments, North Ryde, NSW, Australia). This was then recorded using a custom LabVIEW program (V.12.0f3, National Instruments) and saved offline for secondary processing. Force applied to the system was directly measured using the vertical component of the ground reaction force. The onset of the countermovement was considered as the sample in which system mass reduced $>4 \times SD$ of the preceding system mass (18). This event also represented the point at which integration commenced to establish impulse-, velocity-, and power-time curves. Measures of RFD, impulse, and peak force were calculated across the loaded conditions (i.e., from +20% to +80%). Rate of force development was calculated using the change in force with respect to time between the lowest force value during the countermovement and the force at 200 ms from this point. This method was chosen as fixed time point RFD calculations typically display greater sensitivity to training (14), while 200 ms represented a meaningful proportion of the jump in each condition, is common of many explosive actions in sport (1) and was less than the time it took to achieve peak force. Impulse was calculated as the integral of net force and time from movement onset until toe-off (<20 N). The highest force value during the jump represented peak force.

Force-Velocity Relationship. The product of the force-time and velocity-time curves in all jump conditions generated a series of power-time curves. The force and velocity at peak power was then extracted from each curve and used as x/y coordinates enabling the construction of the force-velocity relationship at the whole system level.

Jump Strategy. The force-time waveform of the unloaded jump squat was used to directly compare the changes in jump strategy. To achieve this, the time delta between each sample was altered to generate 500 samples from the initiation of the countermovement until the subject left the force plate (<20 N) (The Mathworks, Inc., Natick, MA) (8,9). This enabled a direct comparison of the force-time curves at each of the 500 instances.

Statistical Analyses

Statistical analyses on RFD and impulse outcomes, 1RM, peak force, velocity at peak power, and force at peak power for each loading condition were performed in SPSS (SPSS v.24, SPSS Inc., Chicago, IL). The effect of time point (baseline, mid-test, and post-test) was assessed using a one-way repeated-measures analysis of variance (ANOVA) on the sample of stronger athletes and the sample of weaker athletes. Friedman's test was used when data were not normally distributed. The effect of group (stronger and weaker) and time point on the change scores for the above variables were assessed using a two-way ANOVA with repeated measures. The Mann-Whitney *U* test was applied when data were not normally distributed. Post hoc tests using a Bonferroni correction for multiple comparisons was applied when a main significant effect was identified. The magnitude of between-group differences was assessed using Hedges' *g* with confidence intervals bootstrapped to 10,000 iterations (11). The following descriptors were used to interpret these effect size (ES): <0.20 = trivial, 0.21–0.60 = moderate, 0.61–1.20 = large, 1.21–2.0 = very large (13). One-dimensional statistical parametric mapping was applied to evaluate differences between baseline, mid-test, and post-test force-time waveforms across the whole jump (http://www.spm1d.org/index.html; Matlab [Mathworks]). The alpha level indicating a significant effect was set at 0.05 for all analyses.

Results

1 Repetition Maximum Squat

Results showed a significant main effect of time point on 1RM squat in the weaker group ($p = 0.004$). Post hoc tests showed that the weaker subjects displayed significant increases in 1RM squat strength from baseline (1.38 ± 0.32 kg·kg^{-1}) to mid-test (1.49 ± 0.27 kg·kg^{-1}) ($p = 0.009$) and post-test (1.50 ± 0.23 kg·kg^{-1}; $p = 0.021$). A main effect of time point on 1RM was not found in the stronger group (baseline: 2.00 ± 0.16 kg·kg^{-1}; mid-test: 2.03 ± 0.24 kg·kg^{-1}; post-test: 2.06 ± 0.25 kg·kg^{-1}; $p = 0.386$).

Loaded Jump Squat

At mid-test, significant and very large improvements (Figure 1A) were present in RFD at 20% 1RM within the stronger subjects ($p = 0.01$). At this time point, the weaker group displayed significant and moderate improvements (Figure 1C) in RFD at 60% 1RM ($p = 0.01$). The stronger group also displayed significant and very large to large improvements at post-test in this variable at both 20 ($p = 0.05$) and 40% 1RM ($p = 0.03$), respectively (Figure 1A and B). However, only trivial to small, nonsignificant changes were found in the stronger subjects in RFD, impulse or peak force at 60 and 80% 1RM across any 2 time points (Figures 1 and 2). The weaker group significantly improved peak force at 20% ($p = 0.01$) and 60% 1RM at mid-test ($p = 0.045$), alongside impulse at 20% 1RM ($p = 0.008$), all to a small magnitude (Figure 2). At post-test, small but significant improvements (Figure 2A and C) were present in the weaker group in impulse at 20% ($p = 0.02$) and 60% ($p = 0.003$) 1RM. However, the only significant improvements in RFD for the 40% 1RM condition in the weaker group were at post-test vs. baseline ($p = 0.048$, moderate effect) and post-test vs. mid-test ($p = 0.014$, moderate effect) (Figure 1B). None of the changes were significantly different between groups. A comparison of performance measures for both groups throughout the training intervention are presented in Table 1.

Force-Velocity Relationship

At post-test, the stronger subjects displayed a significant upward shift at both the higher velocity and higher force regions of the force-velocity curve (Figure 3). Conversely, a significant rightward translation at the higher force portion of the curve combined with an upward shift under higher velocity conditions was present within the weaker group (Figure 3). No significantly different changes existed between groups.

Jump Strategy

The stronger group showed a significant change in the force-time waveform from baseline to post-test between 50 and 63% of the CMJ ($p < 0.05$) (Figure 4). The baseline to mid-test force-time waveforms and the mid-test to post-test force-time waveforms were not significantly different in the stronger group.

The weaker group showed significant changes in the force-time waveform from (a) baseline to mid-test between 20 and 23%, (b) mid-test to post-test between 44 and 53%, and (c) baseline to post-test between 17 and 26% and between 44 and 61% ($p < 0.05$) of the CMJ (Figure 5).

Discussion

The primary objective of this investigation was to explore adaptations in the timing and magnitude of force production in loaded conditions after weightlifting derivative-based training in subjects with a higher or lower weightlifting ability. Although no statistically significant between-group differences in adaptations were present, the ES comparisons revealed marked early-stage improvements in RFD at higher velocities in the stronger group, while the weaker subjects displayed more notable changes in higher force conditions after 10 weeks.

The current results show that very large (ES: 2.10) changes in RFD at 20% 1RM after only 5 weeks of training, which is in contrast to the small (ES: 0.27) and moderate (ES: 0.71) changes experienced in the weaker group at mid-test and post-test,

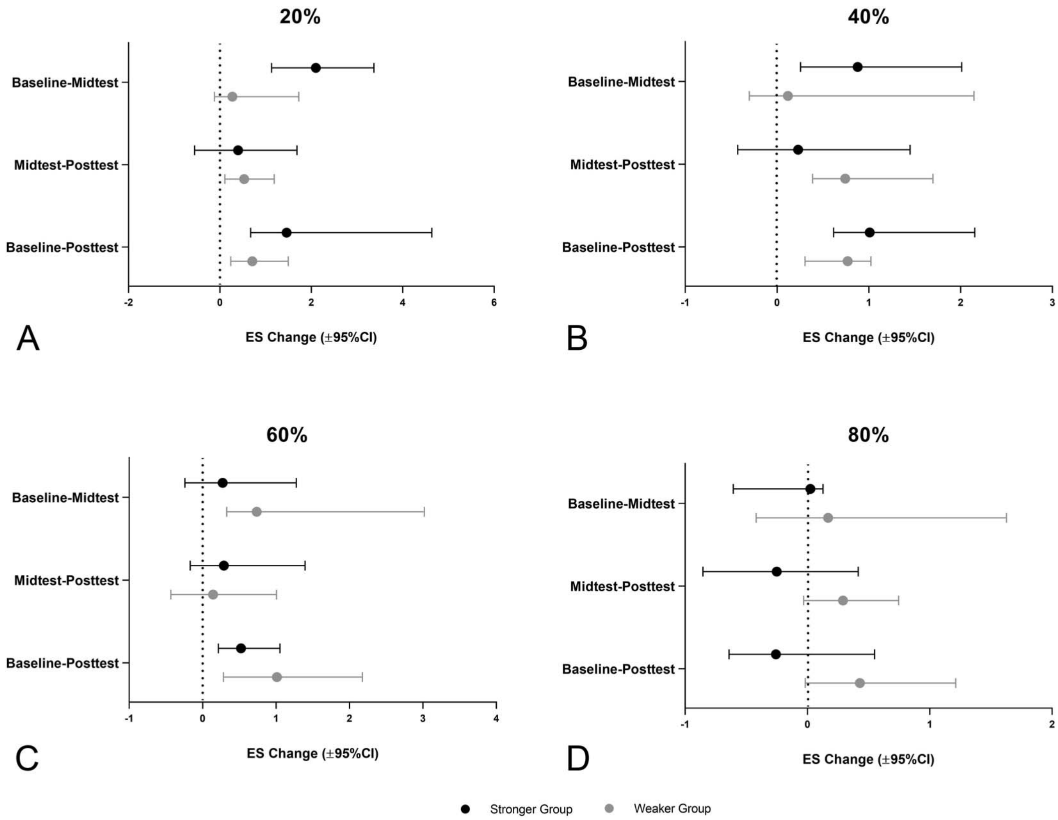

Figure 1. *Hedge's g* effect size (ES ± 95% confidence intervals) changes in rate of force development between time points in each loading condition (% squat 1RM) for the jump squat. Black = stronger group; gray = weaker group. RM = repetition maximum. A = 20% 1RM; B = 40% 1RM; C = 60% 1RM; D = 80% 1RM.

respectively. Consequently, these findings will be of practical relevance to sports scientists and strength coaches. Such an outcome indicates a superior ability to improve the timing of force production in higher velocity conditions in those with a greater pre-existing strength level. However, the magnitude of force production, as measured by peak force, across both heavy and light conditions was similar between both groups. A number of strength qualities underpin performance in the weightlifting derivatives. These qualities include high RFD, the ability to use the stretch-shortening cycle, high maximal strength, high-velocity strength, and high segmental coordination (25). Because maximal strength is already high in the stronger group (1RM squat: stronger = 2.0 kg·kg^{-1}; weaker = 1.38 kg·kg^{-1}), it can be expected that changes stemming from this training may have a greater impact on the temporal aspects, rather than the magnitude of force production (26). This is supported by previous literature that indicated that individuals with greater levels of muscular strength may need a novel training stimulus to continue to adapt (19,26,27) and, in effect, "tuning" the neuromuscular system to use the preexisting force capabilities more rapidly (4). In most cases, the cause of higher levels of strength is due to long-term participation in resistance training activities (as evidenced by the difference in training experience between groups in this study), and as such, these findings are in support of the "window of adaptation" notion, which suggests that as a quality becomes more developed, further adaptations are progressively harder to achieve (21,30). Accordingly, those who are stronger are more likely to enhance the rate of force application than they are peak force itself in higher velocity conditions. By contrast, those who are less trained can expect more general adaptations across all functions that drive maximal neuromuscular expressions (i.e., strength) (9,18,21). Although these findings describe the impact of baseline explosive strength levels on training adaptations, it remains unclear what effect initial strength capabilities have on performance changes if training experience is controlled for.

The present training also resulted in practically relevant differences in the magnitude of adaptation in RFD under the heavier conditions (+60% and +80% 1RM) between groups. At the +60% 1RM jump load, the weaker subjects displayed improvements approximately twice that of their stronger counterparts after 10 weeks (weaker ES = 1.01; stronger ES = 0.52). Similarly, after training, the weaker group displayed an enhancement at 80% 1RM of ES = 0.43 while the stronger group experienced a small loss in this function (ES = −0.26). Therefore, although no statistically significant differences existed between groups in the adaptations experienced, these results are of considerable relevance in practical settings.

Despite the training intervention using relative loads based off the individual's strength level, these findings highlight the distinctly different responses to a weightlifting-based training intervention on the basis of weightlifting ability. Of note, those who were less trained improved their capacity to apply force rapidly under very heavy conditions and therefore provide further

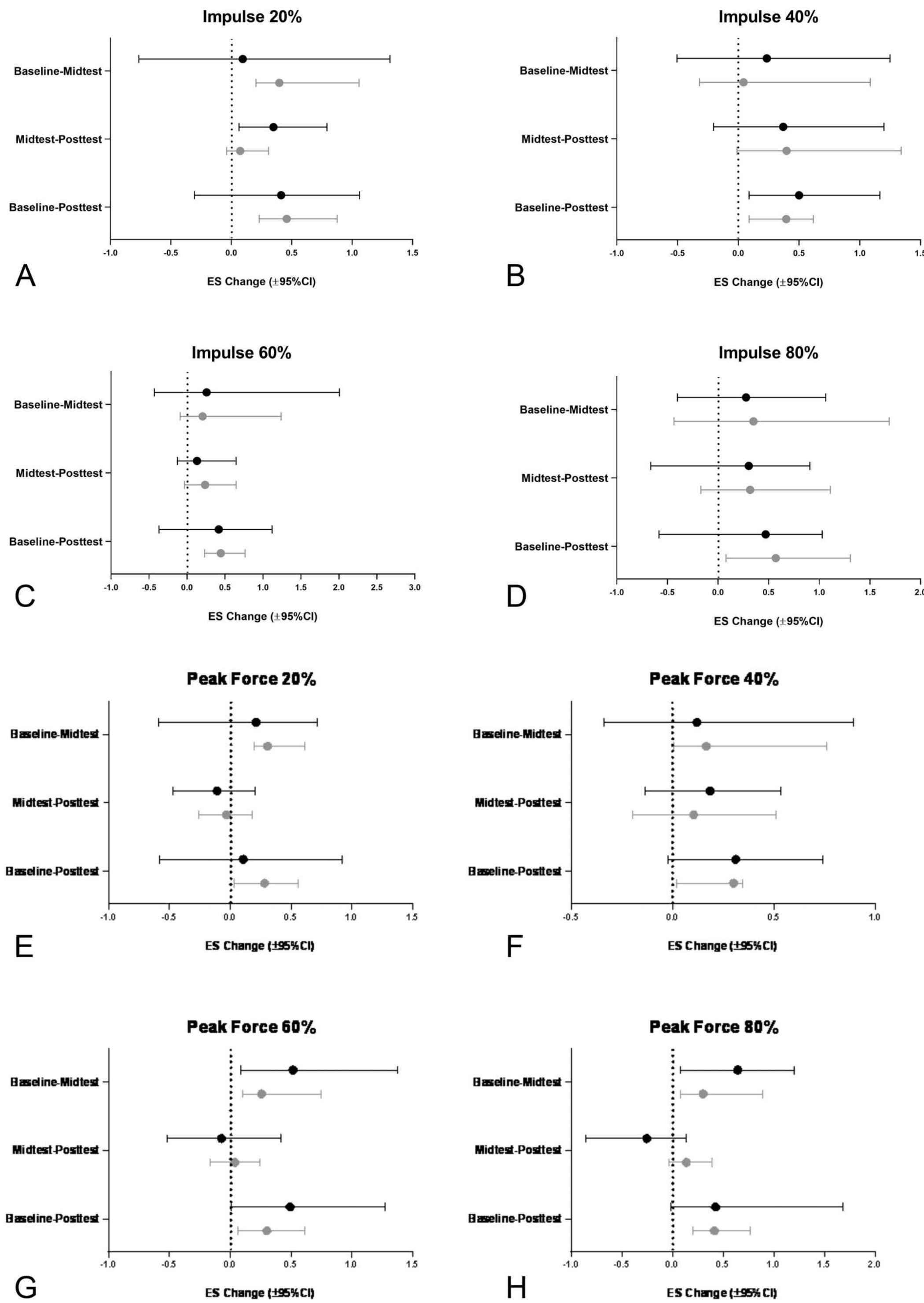

Figure 2. Hedge's *g* effect size (ES) changes in impulse and peak force between time points in each loading condition (% squat 1RM) for the jump squat. Black = stronger group; gray = weaker group. RM = repetition maximum.

evidence of the benefits of such training in those with less experience (17). This training may also be sufficient to increase strength in weaker subjects as evidenced by their significant increase in the 1RM squat. However, improvements in loaded jump squat performance may take much longer when compared with stronger individuals, which is likely a consequence of stronger individuals possessing the neuromuscular characteristics (motor unit recruitment, rate coding, motor unit synchronization, and neuromuscular inhibition) (26) that are required to develop rapid expressions of force. Therefore, it is likely that progressive gains would still be made by the weaker subjects if training continued beyond the 12-week period examined. The limited late-stage

Table 1

Comparison of performance measures for both groups throughout the 10-week training intervention.

	Stronger			Weaker		
	Baseline	Mid-test	Post-test	Baseline	Mid-test	Post-test
Rate of force development (N·s^{-1})						
20%	2,435 ± 533	3,771 ± 662*	4,288 ± 1,609*	2,164 ± 1,980	2,668 ± 1,502	3,689 ± 2,086
40%	3,268 ± 1,298	4,604 ± 1,563	5,034 ± 1,944*	2,791 ± 2,156	3,036 ± 1,689	4,383 ± 1,734*†
60%	3,450 ± 1,927	3,970 ± 1,652	4,530 ± 1,977	2,378 ± 2,281	4,142 ± 2,239*	4,430 ± 1,471
80%	3,537 ± 1,797	3,588 ± 2,235	3,053 ± 1,762	2,584 ± 2,410	2,977 ± 1,966	3,594 ± 2,034
Impulse (N·s^{-1})						
20%	220.1 ± 22.6	222.3 ± 19.0	229.4 ± 19.5	189.3 ± 41.3	206.7 ± 41.1*	210.0 ± 43.9*
40%	224.4 ± 29.6	230.6 ± 18.7	241.3 ± 34.2	201.1 ± 41.5	202.8 ± 31.1	220.4 ± 50.2
60%	212.9 ± 16.4	219.5 ± 29.9	223.8 ± 30.6	188.2 ± 43.6	197.6 ± 43.5	208.6 ± 43.0*
80%	159.0 ± 31.0	171.2 ± 49.9	197.7 ± 105.5	153.0 ± 43.1	166.0 ± 24.7	174.6 ± 26.2
Peak force (N)						
20%	1,994 ± 125	2,020 ± 107	2,007 ± 113	1,795 ± 336	1,903 ± 326*	1,893 ± 317
40%	2,280 ± 137	2,297 ± 133	2,322 ± 116	2,050 ± 374	2,119 ± 406	2,159 ± 302
60%	2,530 ± 145	2,612 ± 156	2,601 ± 129	2,212 ± 418	2,323 ± 395*	2,338 ± 367
80%	2,778 ± 177	2,889 ± 144	2,850 ± 142	2,383 ± 429	2,509 ± 351	2,563 ± 390

*Significant change from baseline.
†Significant change from mid-test.

improvements in the stronger group is in alignment with previous research (18) indicating that even with a mixed-method power training intervention, greater variation is needed to continue to promote adaptations when one is stronger (and therefore more experienced in resistance training). It should also be noted that the inability for this present training intervention to increase maximal strength (as measured by the 1RM squat) in the stronger group is likely responsible for the lack of improvement in RFD under the heavier loads. From a practical perspective, this indicates that even when heavy loads are included in an entirely ballistic resistance training intervention, traditional heavy strength training should not be removed if adaptations are to be maximized in stronger individuals. However, this is based on the assumption that strength level is related to training experience. It may be such that inherently strong individuals with limited training experience are less likely to display diminished returns and therefore require less variation than those with a similar strength level and greater training age.

The training program within the current study used a combined loading approach where both force (i.e., heavier loading) and velocity stimuli (i.e., lighter loads) were included. It is believed that this method will allow an individual to increase both their force and velocity production characteristics (10,21,24). This notion is supported in part by the findings of the present investigation whereby training produced changes in the weaker subjects' force-velocity profile by shifting the entire profile up (e.g., greater velocities) and to the right (e.g., greater force production) (Figure 3). Similar results were also displayed by Cormie et al. (5) who showed that a strength-power (combined loading) program produces greater adaptations across a loading

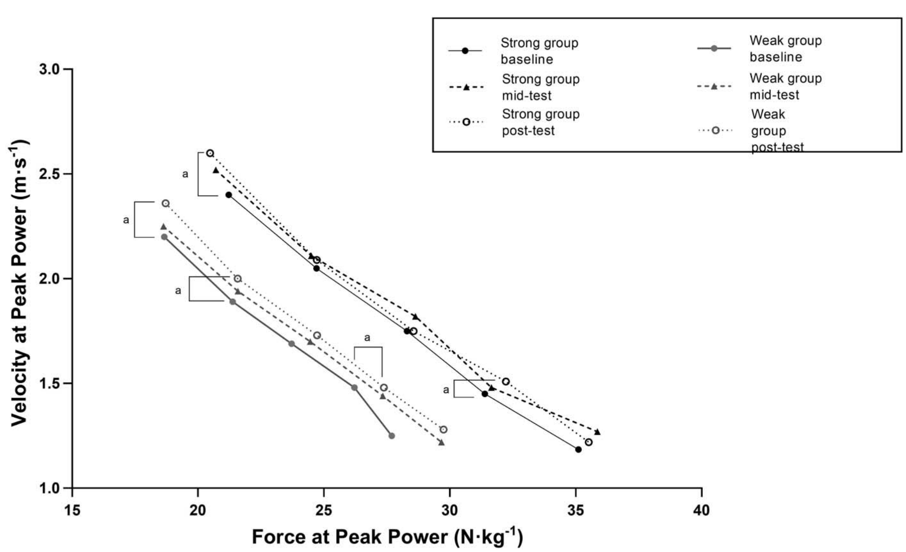

Figure 3. Changes in the force-velocity relationship derived from the incremental load jump squat between time points. Black = stronger group; gray = weaker group. a = significant change between time points. Changes in the force contributions to peak power occur horizontally; changes in the velocity contributions to peak power occur vertically.

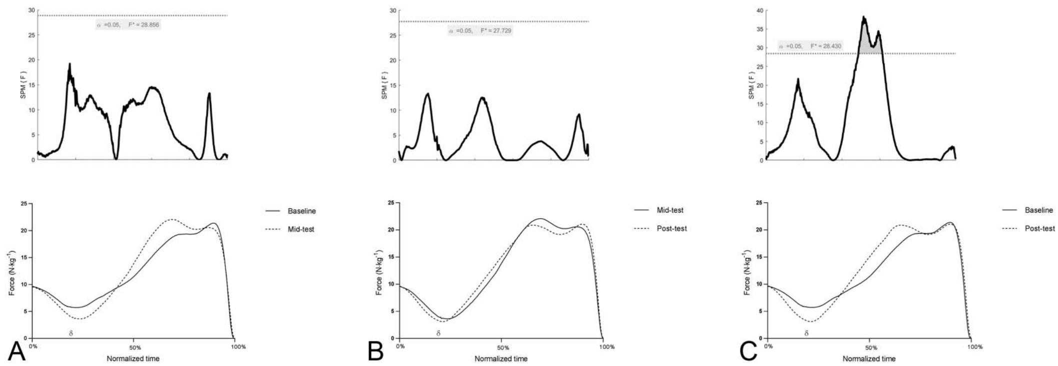

Figure 4. Comparison of the force-time curves for the stronger group between baseline and mid-test (A), mid-test to post-test (B), and baseline to post-test (C). The SPM [F] values in the upper plots represent the F-statistic and are plotted as a function of time for each comparison. The horizontal dotted lines indicate the critical threshold. Differences were deemed significant when SPM [F] exceeded the critical threshold. SPM = statistical parametric mapping.

spectrum compared with power program. Considering that the product of force and velocity is power, it seems that a combined-method approach that uses weightlifting derivatives may allow individuals to improve their power output, which has been considered by many as an important performance characteristic (2,7,23). Although the current study provides evidence that a combined loading approach may be effective for both stronger and weaker individuals, it seems that stronger individuals with greater training experience may require a unique loading stimulus to allow them to continue to adapt beyond 5 weeks.

An examination of the CMJ force-time waveform provides insight into the strategy of force application throughout the entire movement. Previous research has revealed that such an analysis is sensitive to detect changes in jump strategy after training (9,18), and to distinguish between athletes of different competition grades (20). The results of the current study suggest that both stronger and weaker subjects changed their jump strategy after each training phase. As indicated in Figures 4 and 5, both groups of subjects developed a more rapid unweighting phase that ultimately resulted in large increases in their eccentric RFD. This adaptation was more clearly demonstrated by the weaker group most likely due to the greater window available for adaptation (19,26,27). These collective findings, alongside those of previous ballistic strength and power training interventions (6,18), suggest that both stronger and weaker subjects increased their ability to use the stretch-shortening cycle. Specifically, subjects were able to tolerate a greater eccentric demand in the form of a more rapid eccentric RFD and were then able to effectively transition into the concentric (propulsion) phase to produce greater forces and velocities under various loads.

Taken together, these findings reveal that changes in the rate and timing of maximal force production stemming from weightlifting training seem to be dependent on weightlifting ability. Stronger individuals likely experience very large improvements in RFD under lighter loads in the early stages of training, with limited further improvements after 5 weeks. Despite the inclusion of heavy loads, these subjects also displayed a decay in force-producing capabilities under heavy conditions. Notably, different adaptations were experienced within weaker subjects including improvements in both heavy and lighter measures of RFD after 10 weeks of training, with only minor changes occurring at 5 weeks. However, it is important to note the confounding factor of training experience resulting in stronger individuals displaying a reduced potential for adaptation due to a greater training age. Further research should look to explore this factor to better understand adaptations to strength-power training.

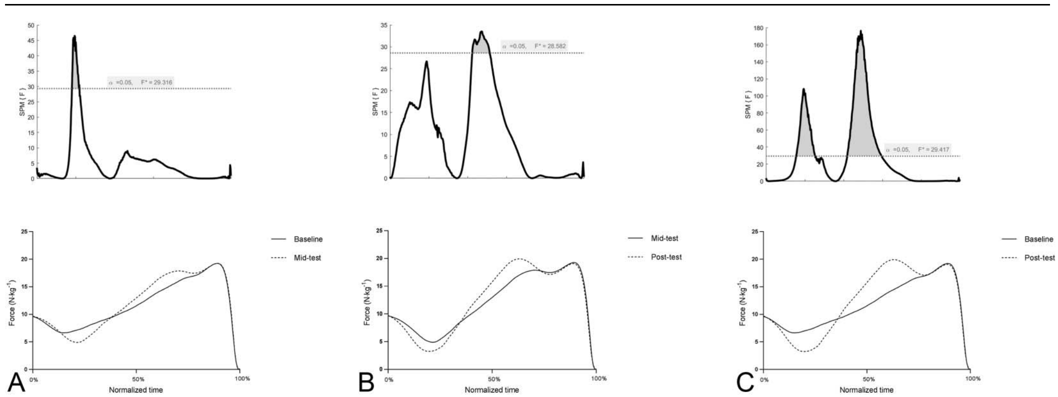

Figure 5. Comparison of the force-time curves for the weaker group between baseline and mid-test (A), mid-test to post-test (B), and baseline to post-test (C). The SPM [F] values in the upper plots represent the F-statistic and are plotted as a function of time for each comparison. The horizontal dotted lines indicate the critical threshold. Differences were deemed significant when SPM [F] exceeded the critical threshold. SPM = statistical parametric mapping.

Practical Applications

Because RFD is a vital performance characteristic, these findings can be used by sports scientists and strength coaches to inform the design of resistance training programs. Important practical considerations include (a) Individuals stronger in the weightlifting derivatives improve high-velocity RFD at a faster rate and to a greater extent after training with these lifts. However, these improvements will be short-lived, likely because of the absence of other resistance training stimuli (i.e., heavy strength training); (b) those who are less proficient will experience improvements in RFD in both heavy and lighter conditions; however, these changes occur over a longer time course.

References

1. Aagaard P, Simonsen E, Andersen J, Magnusson P, Dyhre-Poulsen P. Increased rate of force development and neural drive of human skeletal muscle following resistance training. *J Appl Physiol* 93: 1318–1326, 2002.
2. Baker D. A series of studies on the training of high-intensity muscle power in rugby league football players. *J Strength Cond Res* 15: 198–209, 2001.
3. Baker D. Differences in strength and power among junior-high, senior-high, college-aged, and elite professional rugby league players. *J Strength Cond Res* 16: 581–585, 2002.
4. Bobbert MF, Van Soest AJ. Effects of muscle strengthening on vertical jump height: A simulation study. *Med Sci Sports Exerc* 26: 1012, 1994.
5. Cormie P, McCaulley GO, McBride JM. Power versus strength-power jump squat training: Influence on the load power relationship. *Med Sci Sports Exerc* 39: 996–1003, 2007.
6. Cormie P, McGuigan M, Newton R. Changes in the eccentric phase contribute to improved stretch-shorten cycle performance after training. *Med Sci Sports Exerc* 42: 1731–1744, 2010.
7. Cormie P, McGuigan M, Newton R. Developing maximal neuromuscular power: Part 2 -training considerations for improving maximal power production. *Sports Med* 41: 125–146, 2011.
8. Cormie P, McGuigan MR, Newton RU. Adaptations in athletic performance after ballistic power versus strength training. *Med Sci Sports Exerc* 42: 1582, 2010.
9. Cormie P, McGuigan MR, Newton RU. Influence of strength on magnitude and mechanisms of adaptation to power training. *Med Sci Sports Exerc* 42: 1566–1581, 2010.
10. Haff GG, Nimphius S. Training principles for power. *Strength Cond J* 34: 2–12, 2012.
11. Hentschke H, Stüttgen MC. Computation of measures of effect size for neuroscience data sets. *Eur J Neurosci* 34: 1887–1894, 2011.
12. Hoffman JR, Cooper J, Wendell M, Kang J. Comparison of Olympic vs. traditional power lifting training programs in football players. *J Strength Cond Res* 18: 129–135, 2004.
13. Hopkins W, Marshall S, Batterham A, Hanin J. Progressive statistics for studies in sports medicine and exercise science. *Med Sci Sports Exerc* 41: 3–13, 2009.
14. Ishøi L, Clausen MB, Aagaard P. Inappropriate methods and flawed conclusion in: Can resistance training enhance the rapid force development in unloaded dynamic isoinertial multijoint movements? A systematic review. *J Strength Cond Res* 32: e1–e2, 2018.
15. James L, Haff GG, Kelly V, Beckman E. Towards a determination of the physiological characteristics distinguishing successful mixed martial arts athletes: A systematic review of combat sport literature. *Sports Med* 46: 1525–1551, 2016.
16. James LP, Beckman EM, Kelly VG, Haff GG. The neuromuscular qualities of higher and lower-level mixed martial arts competitors. *Int J Sports Physiol Perform* 12: 612–620, 2017.
17. James LP, Comfort P, Suchomel TJ, et al. The impact of power clean ability and training age on adaptations to weightlifting-style training. *J Strength Cond Res* 33: 2936–2944, 2019.
18. James LP, Haff GG, Kelly VG, et al. The impact of strength level on adaptations to combined weightlifting, plyometric, and ballistic training. *Scand J Med Sci Sports* 28: 1494–1505, 2018.
19. Kraemer WJ, Newton RU. Training for muscular power. *Phys Med Rehabil Clin* 11: 341–368, 2000.
20. McMahon JJ, Murphy S, Rej SJ, Comfort P. Countermovement jump phase characteristics of senior and academy rugby league players. *Int J Sports Physiol Perform* 12: 803–811, 2017.
21. Newton RU, Kraemer WJ. Developing explosive muscular power: Implications for a mixed methods training strategy. *Strength Cond J* 16: 20–31, 1994.
22. Stone MH, Moir G, Glaister M, Sanders R. How much strength is necessary? *Phys Ther Sport* 3: 88–96, 2002.
23. Stone MH, O'Bryant HS, McCoy L, et al. Power and maximum strength relationships during performance of dynamic and static weighted jumps. *J Strength Cond Res* 17: 140–147, 2003.
24. Suchomel TJ, Comfort P, Lake JP. Enhancing the force–velocity profile of athletes using weightlifting derivatives. *Strength Cond J* 39: 10–20, 2017.
25. Suchomel TJ, Comfort P, Stone MH. Weightlifting pulling derivatives: Rationale for implementation and application. *Sports Med* 45: 823–839, 2015.
26. Suchomel TJ, Nimphius S, Bellon CR, Stone MH. The importance of muscular strength: Training considerations. *Sports Med*: 1–21, 2018.
27. Suchomel TJ, Nimphius S, Stone MH. The importance of muscular strength in athletic performance. *Sports Med* 46: 1419–1499, 2016.
28. Tidow G. Aspects of strength training in athletics. *New Stud Athletics* 1: 93–110, 1990.
29. Tricoli V, Lamas L, Carnevale R, Ugrinowitsch C. Short-term effects on lower-body functional power development: Weightlifting vs. vertical jump training programs. *J Strength Cond Res* 19: 433–437, 2005.
30. Wilson G, Murphy A, Walshe A. Performance benefits from weight and plyometric training: Effects of initial strength level. *Coaching Sport Sci J* 2: 3–8, 1997.